



Department of Computer Science and Business Systems

Academic Year 2024–2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech CSBS

Course Code & Title: CS3391 Object Oriented Programming

Name of the Faculty member: Dr.S.Erana Veerappa Dinesh, ASP/CSBS

Innovative Practice Description

- **Unit/Topic:** Unit 4 / String Methods
- **Course Outcome:** The students will be able to make use of string handling methods to solve real-world problems
- **Topic Learning Outcome:** TLO11
- **Activity Chosen:** Crossword Puzzle
- **Justification:**
Various String methods in Java can be easily paired with the help of crossword puzzle which leads to the greater retention of the data among the students.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - A crossword puzzle is an engaging word game that encourages students to recall concepts and independently find solutions. Unlike traditional MCQ tests, it stimulates curiosity and interest among learners.
 - This activity was conducted in a classroom over 20 minutes, focusing on the topic of built-in exceptions in Java. Students were presented with a crossword puzzle containing 10 questions, which they had to solve by thinking critically and writing the answers in the puzzle grid.
 - The activity took place during the last 20 minutes of the class, followed by a review of the answers. Sample solutions were illustrated in Figure 1. Incorrect answers were discussed at the end of the session, ensuring all students understood the correct responses to every question.

CO–PO/ PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	3	3	3	1	1	1	3	1	1

(1– Low 2 –Moderate 3 – High)

• PO/ PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1
	3	3	3	3
Justification for correlation	The students apply the knowledge of string methods in java	The students analyze the given questions and identify solutions	The students find the solutions for the given question.	The students apply the methods of strings for solving the novel problems

• Images/Screenshot of the practice:

String Methods in Java

M. Esakkiammal
953623244009

20/10/24

ACROSS

2. returns the canonical representation of the string
4. returns the position of the specified character in the string
7. compares two strings in the dictionary order
8. replace the first matching substring
10. breaks the string into an array of strings
11. converts the string to an array of bytes

DOWN

1. checks whether the string is equal to charSequence
3. converts the string to a char array
5. returns a subsequence from the string
6. returns the substring of the string
9. returns the string representation of the specified argument
12. removes any leading and trailing whitespaces

String Methods in Java

S. Rajakaleeswaran

953623244037

ACROSS

2. returns the canonical representation of the string
4. returns the position of the specified character in the string
7. compares two strings in the dictionary order
8. replace the first matching substring
10. breaks the string into an array of strings
11. converts the string to an array of bytes

DOWN

1. checks whether the string is equal to charSequence
3. converts the string to a char array
5. returns a subsequence from the string
6. returns the substring of the string
9. returns the string representation of the specified argument
12. removes any leading and trailing whitespaces

Fig1: Sample Answers by Esakkiammal M and Rajakaleeswaran S of II year CSBS

• Reflective Critique:

✓ *Feedback of practice from students and other stakeholders:*

- Some of the students find it very easy to identify the various methods used in java string handling methods.
- Some of them take more time to complete the puzzle and need more clarification on the methods in strings.

✓ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to think on their own and develop the java programs using string handling methods. The students level of understanding on this topic gets increased.

✓ *Challenges faced in implementation:*

- The students who didn't understand the concept clearly were unable to complete it within the allotted time.
- Few of them were struggled to find the answer for the questions.

- **References:**

- ❖ https://www.ritrjpm.ac.in/images/CSBS/2023-2024/TPS_GE3151_PSPP_BY_23_24.pdf
- ❖ https://www.ritrjpm.ac.in/images/CSBS/2022-2023/SEV_GE3151_CrosswordPuzzle.pdf
- ❖ <http://www.classtools.net/crossword/>
- ❖ <https://www.education.com/worksheet-generator/reading/crossword-puzzle/>